

**FIITJEE**  
**ALL INDIA TEST SERIES**  
**JEE (Advanced)-2024**  
**PART TEST – I**  
**PAPER –2**  
**TEST DATE: 19-11-2023**

**Time Allotted: 3 Hours**

**Maximum Marks: 180**

**General Instructions:**

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.  
**Section – A (01 – 04, 18 – 21, 35 – 38):** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.  
**Section – A (05 – 07, 22 – 24, 39 – 41):** This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).  
**Section – B (08 – 13, 25 – 30, 42 – 47):** This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.  
**Section – C (14 – 17, 31 – 34, 48 – 51):** This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

**MARKING SCHEME**

**Section – A (Single Correct):** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                                                     |
|----------------|---|----|---------------------------------------------------------------------|
| Full Marks     | : | +3 | If ONLY the correct option is chosen.                               |
| Zero Marks     | : | 0  | If none of the options is chosen (i.e. the question is unanswered); |
| Negative Marks | : | -1 | In all other cases.                                                 |

**Section – A (One or More than One Correct):** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                                                                                     |
|----------------|---|----|-----------------------------------------------------------------------------------------------------|
| Full Marks     | : | +4 | If only (all) the correct option(s) is (are) chosen;                                                |
| Partial Marks  | : | +3 | If all the four options are correct but ONLY three options are chosen;                              |
| Partial marks  | : | +2 | If three or more options are correct but ONLY two options are chosen and both of which are correct; |
| Partial Marks  | : | +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option;        |
| Zero Marks     | : | 0  | If none of the options is chosen (i.e. the question is unanswered);                                 |
| Negative Marks | : | -2 | In all other cases.                                                                                 |

**Section – B:** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                         |
|----------------|---|----|-----------------------------------------|
| Full Marks     | : | +4 | If ONLY the correct integer is entered; |
| Zero Marks     | : | 0  | Question is unanswered;                 |
| Negative Marks | : | 0  | In all other cases.                     |

**Section – C:** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                         |
|----------------|---|----|-----------------------------------------|
| Full Marks     | : | +3 | If ONLY the correct integer is entered; |
| Zero Marks     | : | 0  | Question is unanswered;                 |
| Negative Marks | : | 0  | In all other cases.                     |

## Physics

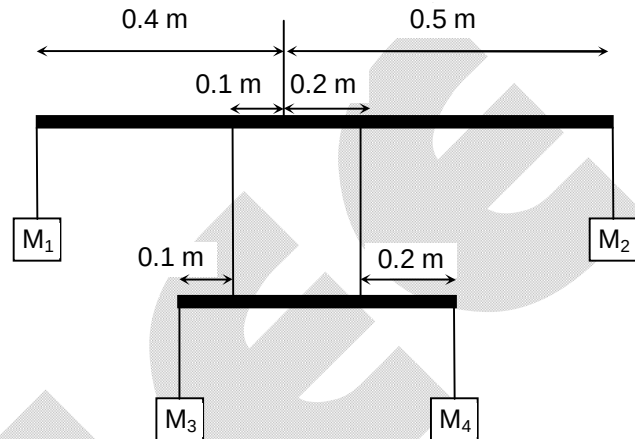
### PART – I

#### SECTION – A

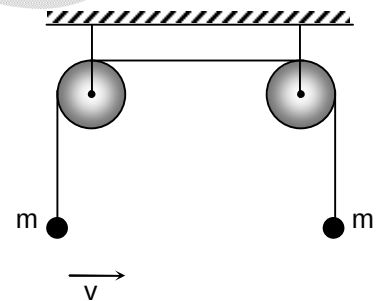
(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

1. In the diagram below, the two cross beams and the seven supporting strings are all massless. The hanging objects are  $M_1 = 400$  g,  $M_2 = 200$  g, and  $M_4 = 500$  g. What is the value of  $M_3$  for the system to be in static equilibrium?
- (A) 400 g  
 (B) 500 g  
 (C) 600 g  
 (D) 700 g



2. Two particles of mass  $m$  are connected by pulleys as shown. The mass on the left is given a small horizontal velocity, and oscillates back and forth. The mass on the right
- (A) remains at rest  
 (B) oscillates vertically, and with a net upward motion  
 (C) oscillates vertically, and with a net downward motion  
 (D) oscillates vertically, with no net motion



3. Two particles with mass  $m_1$  and  $m_2$  are connected by a massless rigid rod of length  $L$  and placed on a horizontal frictionless table. At time  $t = 0$ , the first mass receives an impulse perpendicular to the rod, giving it speed  $v$ . At this moment, the second mass is at rest. The time after which the second mass is at rest again is

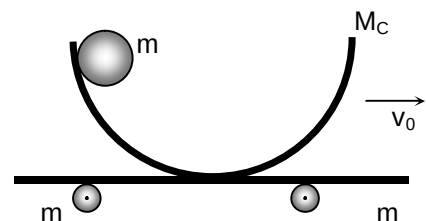
(A)  $t = \frac{2\pi L}{v}$

(B)  $t = \frac{\pi(m_1 + m_2)L}{m_2 v}$

(C)  $t = \frac{2\pi m_2 L}{(m_1 + m_2)v}$

(D)  $t = \frac{2\pi m_1 L}{(m_1 + m_2)v}$

4. A ball of mass  $m$  rolls in a bowl bolted on a cart. The bowl and cart move together and have a combined mass  $M_c$ , as shown in the figure. Initially, the cart moves to the right with speed  $v_0$  with respect to the ground, while the ball is at rest with respect to the cart. The ball is released from the top of the bowl. When it reaches the bottom of the bowl, it moves with velocity  $v_b$  with respect to the cart. At this instant, how fast is the cart moving with respect to the ground?



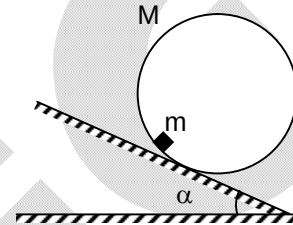
- (A)  $v_0 - v_b$  (B)  $v_0 - \frac{m}{M_c} v_b$   
 (C)  $\frac{M_c v_0 - m v_b}{m + M_c}$  (D)  $\frac{(m + M_c) v_0 - m v_b}{m + M_c}$

### SECTION – A

(One or More than one correct type)

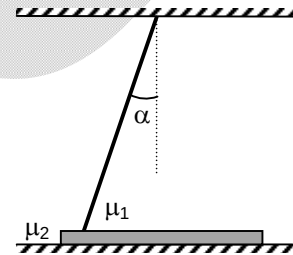
This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. On an incline with slope angle  $\alpha$  there lies a cylinder with mass  $M$ , its axis being horizontal. A small block with mass  $m$  is placed inside it. The coefficient of friction between the block and the cylinder is  $\mu$ ; the incline is nonslippery. What can be the possible slope angle  $\alpha$  for the cylinder to stay at rest. The block is much smaller than the radius of the cylinder.  $\mu = \frac{3}{4}$ ,  $m = 1\text{kg}$ ,  $M = \frac{1}{5}\text{kg}$ .



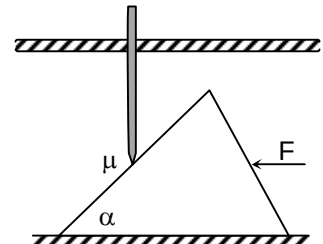
- (A)  $10^\circ$  (B)  $15^\circ$   
 (C)  $40^\circ$  (D)  $60^\circ$

6. A rod with length  $l = 5\text{m}$  is hinged to a ceiling with height  $h = 4\text{m}$ . Underneath, a board is being dragged on the floor. The rod is meant to block the movement the board in one direction while allowing it move in the opposite direction. The coefficient of friction is  $\mu_1$  between the board and the rod, and  $\mu_2$  between the board and the floor. What condition should be fulfilled for it to do its job?



- (A)  $\mu_1 = 0.5, \mu_2 = 0.6$  (B)  $\mu_1 = 0.7, \mu_2 = 0.8$   
 (C)  $\mu_1 = 0.8, \mu_2 = 0.6$  (D)  $\mu_1 = 0.9, \mu_2 = 0.8$

7. A wedge with the angle  $\alpha = 45^\circ$  at the tip is lying on the horizontal floor. There is a hole with smooth walls in the ceiling. A light rod has been inserted snugly into that hole, and it can move up and down without friction, while its axis remains vertical. The rod is supported against the wedge. Assume that the only point which is rough is the contact point of the wedge and the rod. The friction coefficient at the point of contact is  $\mu$ . For which values of  $\mu$  it is possible to push the wedge through, behind the rod, by only applying a sufficiently large horizontal force?



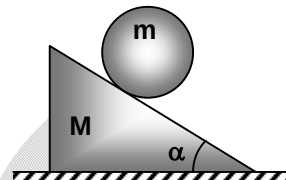
- (A) 0.5 (B) 0.6  
 (C) 0.7 (D) 0.8

**SECTION – B**

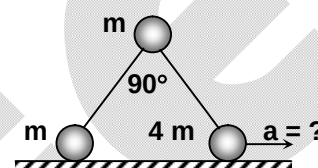
**(Numerical Answer Type)**

This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

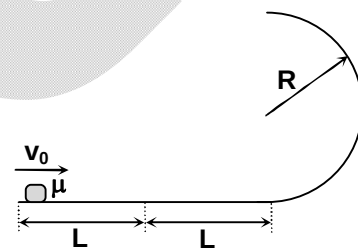
8. A wedge of mass  $M = 10 \text{ kg}$  and inclination  $\alpha = 30^\circ$  is free to move on a smooth horizontal plane. A uniform solid cylinder of mass  $m = 20 \text{ kg}$  is placed on the rough inclined face of the wedge. Find the acceleration (in  $\text{m/s}^2$ ) of the centre of the cylinder relative to the wedge down the inclined face. There is no slipping between the cylinder and the wedge.



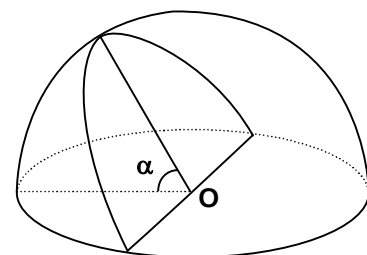
9. Three little cylinders are connected with weightless identical rods, where there is a hinge near the middle cylinder, so that the angle between the rods can change freely. Initially this angle is a right angle. Two of the cylinders have mass  $m$ , another one at the side has the mass  $4m$  as shown in the figure. Find the acceleration of the heavier (in  $\text{m/s}^2$ ) cylinder immediately after the motion begins. Ignore friction. Take  $g = 9 \text{ m/s}^2$ .



10. A small object is initially at rest at the starting point of a horizontal straight track of length  $2L$ . The speed with which it should be pushed in order that after sliding along the vertical semicircular path of radius  $R$  at the end of the straight track it hit the midpoint of the horizontal track is  $v$ . Find value of  $\frac{5v}{21}$ . The coefficient of kinetic friction along the horizontal track is  $\mu$ , the circular track is frictionless. ( $L=2 \text{ m}$ ;  $R=0.5 \text{ m}$ ;  $\mu=0.4$ ,  $g = 10 \text{ m/s}^2$ )



11. A part of the surface of hemisphere is isolated from hemisphere by cutting it by two planes passing through the same diameter and forming an angle with each other. The shape thus obtained is called a "mericarp". The left part is referred as mericarp with central angle  $\alpha$ . If mericarp is cut from solid hemisphere then it is called a solid mericarp. If mericarp is obtained from hemispherical shell it is called a hollow mericarp. Find the distance of centre of mass of a uniform solid mericarp of radius  $R = \frac{16}{\pi}$  and central angle  $\alpha = 10^\circ$ , from the centre of sphere. For angles less than 7 degrees you may assume  $\sin \theta = \theta$ .



12. Find the distance of centre of mass of a uniform hollow mericarp of radius  $R = \frac{16}{\pi}$  and central angle  $\alpha = 10^\circ$ , from the centre of sphere. For angles less than 7 degrees you may assume  $\sin \theta = \theta$ .
13. At what speed (in  $\text{m/s}$ ) must a pebble be thrown from a height of  $h = 3.6 \text{ m}$ , and at an angle of  $30^\circ$ , measured from the horizontal, if it is to hit the ground at an angle of  $45^\circ$ . (Air drag can be neglected.) (Take  $g = 10 \text{ m/s}^2$ )

**SECTION – C**  
(Numerical Answer Type)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

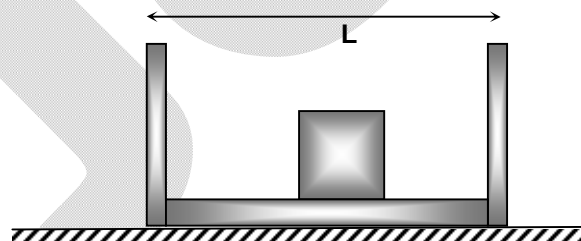
**Paragraph for Question Nos. 14 and 15**

The 15.2 km M4 metro line connects Delhi railway station and Gurugram railway station, and between them it has 7 other stops. Trains leave each station (or stop) with a constant acceleration of  $1.0 \text{ m/s}^2$ , move with constant speed for some time and stop at next station with the same deceleration when brakes are applied. The maximum speed between stations is  $72 \text{ km/h}$ . At the stations the average time to change passengers is 0.5 minutes.

14. During one journey, how long (time in seconds) does the train travel at the speed of  $72 \text{ km/h}$ ?
15. What is the journey time of the M4 metro between the two terminals, i.e. how much time elapses between the departure of a train from Delhi and its arrival at Gurugram railway station?

**Paragraph for Question Nos. 16 and 17**

There is a box A of length  $L = 1.0 \text{ m}$  on the smooth horizontal surface. The center of mass of the box is located at its middle position. Inside the box, there is a block B with negligible size. Box A and block B have the same mass. Initially, box A is at rest and block B is located at the middle of A and moves to the right with an initial speed  $v_0 = 5 \text{ ms}^{-1}$ . The coefficient of friction between A and B is  $\mu = 0.05$  and the collisions of B with left and right walls of A and B are elastic. (Take  $g = 9.8 \text{ m/s}^2$ )



16. How many collisions occur between the walls of box A and block B?
17. What is the horizontal displacement of box A (in metre) from the initial moment to the moment when block B has just reached the relatively stationary position inside the box?

## Chemistry

## PART – II

### SECTION – A

(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

18. Consider the following statements related to the compounds of alkali metals  
 (I)  $\text{Na}_3\text{N}$  is formed by heating sodium with  $\text{N}_2$ .  
 (II)  $\text{CsO}_2$  is paramagnetic and coloured.  
 (III)  $\text{LiCl}$  is deliquescent.  
 (IV)  $\text{LiH}$  is thermally less stable than  $\text{NaH}$ .  
 The option with correct set of statements is  
 (A) (I) and (III) (B) (II) and (IV)  
 (C) (III) and (IV) (D) (II) and (III)
19. The **incorrect** statement among the statements given below is  
 (A) Dipole moment of  $\text{NH}_3$  is greater than that of  $\text{NF}_3$   
 (B) In the solid state of  $\text{KHCO}_3$ ,  $\text{HCO}_3^-$  ions exist as dimeric anion  
 (C) In Borax all the Boron atoms are  $\text{sp}^3$  hybridized  
 (D) In  $\text{SO}_4^{2-}$  all bond lengths are same
20. The compound in which the central atom exhibit same type of hybridization that is shown by Br in  $[\text{BrF}_4]^-$  is  
 (A)  $\text{SF}_2\text{Cl}_2$  (B)  $\text{SF}_4$   
 (C)  $\text{ClF}_3$  (D)  $\text{XeOF}_4$
21. Consider the following statements related to alkaline earth metals and their compounds  
 (I)  $\text{BeO}$  is an amphoteric oxide.  
 (II)  $\text{BaSO}_4$  is more soluble in water than  $\text{CaSO}_4$ .  
 (III)  $\text{BeF}_2$  is soluble in water.  
 (IV)  $\text{Mg}(\text{NO}_3)_2$  decomposes on heating producing  $\text{Mg}(\text{NO}_2)_2$  and  $\text{O}_2$ .  
 The option with correct set of statement is  
 (A) (II) and (IV) (B) (I) and (III)  
 (C) (I) and (IV) (D) (III) and (IV)

### SECTION – A

(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

22. The correct statement(s) related to the equilibrium  

$$\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$$
  
 is/are  
 (A) At equilibrium addition of  $\text{NH}_2\text{COONH}_4(\text{s})$  increases the partial pressure of  $\text{NH}_3$  at constant temperature  
 (B) Addition of  $\text{NH}_3(\text{g})$  at equilibrium decreases the value of  $K_c$ .  
 (C) Addition of inert gas at constant volume does not affect this equilibrium.  
 (D) Addition of catalyst does not affect the equilibrium state.



23. The solution (s) in which the solubility of AgCN will be greater than that in pure water is(are)  
 $[K_{sp}(\text{AgCN}) = 4 \times 10^{-16}, K_a(\text{HCN}) = 5 \times 10^{-10}]$   
 (A) 0.01 M AgNO<sub>3</sub> solution (B) A buffer solution of pH = 11  
 (C) 0.3 M NH<sub>3</sub> solution (D) A buffer solution of pH = 5
24. Rate constant k varies with temperature by equation  
 $\log k(\text{min}^{-1}) = 4 - \frac{3000}{T(\text{K})}$   
 The correct statement(s) related to the above equation is/are [Given : (R = 2 cal mole<sup>-1</sup> K<sup>-1</sup>)]  
 (A) Pre exponential factor A is 4 (B) E<sub>a</sub> is 13.818 Kcal  
 (C) E<sub>a</sub> is 3000 cal (D) When T → ∞, k = 10<sup>4</sup> min<sup>-1</sup>

### SECTION – B

#### (Numerical Answer Type)

*This section contains SIX (06) Numerical based questions. The answer to each question is a NON-NEGATIVE INTEGER VALUE.*

25. Among SF<sub>3</sub>Cl, XeO<sub>2</sub>F<sub>2</sub>, [ICl<sub>4</sub>]<sup>-</sup>, [NH<sub>4</sub>]<sup>+</sup>, [ICl<sub>6</sub>]<sup>-</sup>, XeO<sub>3</sub>, SF<sub>4</sub>, [XeF<sub>5</sub>]<sup>+</sup> and [XeO<sub>6</sub>]<sup>4-</sup>, the total number of species having sp<sup>3</sup>d hybridized central atom is \_\_\_\_\_
26. Among N<sub>2</sub><sup>-</sup>, O<sub>2</sub><sup>2-</sup>, H<sub>2</sub><sup>-</sup>, Li<sub>2</sub>, F<sub>2</sub>, He<sub>2</sub>, N<sub>2</sub><sup>+</sup>, O<sub>2</sub><sup>+</sup>, Be<sub>2</sub> the number of species which are paramagnetic is \_\_\_\_\_
27. Among Zn(OH)<sub>2</sub>, Sn(OH)<sub>2</sub>, Cu(OH)<sub>2</sub>, Fe(OH)<sub>3</sub>, Pb(OH)<sub>2</sub>, Al(OH)<sub>3</sub> the number of hydroxide which are soluble in excess of NaOH solution is \_\_\_\_\_
28. 4.9 mg H<sub>2</sub>SO<sub>4</sub>, 3.65 mg of HCl and 3.15 mg of HNO<sub>3</sub> were taken in a flask of volume 'V' ml. the pH of the final solution is 4.3. The value of  $\frac{V}{100}$  is \_\_\_\_\_ (log5 = 0.7)
29. At 200°C, the gaseous reaction A(g) → 2B(g) + C(g) is found to be of 1<sup>st</sup> order. Starting with pure A(g), if at the end of 5 minutes, the total pressure of the system is 128 mm and after infinite time it is 240 mm and the partial pressure of A(g) at the end of 5 min is 'y' mm. the value of y is \_\_\_\_\_
30. 9 moles of SO<sub>2</sub> reacts completely with acidified potassium dichromate solution. In this reaction the number of moles of potassium dichromate reacted is x and the number of moles of H<sub>2</sub>O produced is y. the value of (x + y) is \_\_\_\_\_

### SECTION – C

#### (Numerical Answer Type)

*This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions of numerical answer type. The answer to each question is a NUMERICAL VALUE (XXXXX.XX). If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.*

#### Paragraph for Question Nos. 31 and 32

The photoelectric work function depends upon the nature of metal. A metal having a higher ionization energy has a greater value of photoelectric work function. The values of photoelectric work functions of Li, Na, K and Cu are 2.42 eV, 2.3 eV, 2.25 eV and 4.8 eV respectively.

(h = 6.6 × 10<sup>-34</sup> Joule sec, 1 eV = 1.6 × 10<sup>-19</sup> J)

31. When sodium metal is exposed to an electromagnetic radiation, the maximum kinetic energy of the photoelectrons was found to be 0.70 eV. The wavelength of the incident radiation is  $y \times 10^{-8}$  m. The value of  $y$  is \_\_\_\_\_
32. An electromagnetic radiation of wavelength  $500 \text{ \AA}$  falls on the surface of copper metal. The maximum kinetic energy of ejected photoelectrons (in eV) is \_\_\_\_\_

**Paragraph for Question Nos. 33 and 34**

$\text{SO}_2$  (g) when circulated through suspension of Colemanite ( $\text{Ca}_2\text{B}_6\text{O}_{11}$ ) powder, compound P and Q are formed. Compound Q on strong heating produced compounds R and S. R is an acidic oxide while S is a neutral oxide. Compound R reacts with  $\text{P}_2\text{O}_5$  to form compound T. The difference in molar mass of P and Q is 140.2. Sum of molar mass of R and S is 87.6.  
[Use molar mass (in  $\text{g mol}^{-1}$ ) : H = 1, B = 10.8, O = 16, S = 32, P = 31]

33. Percentage by mass of oxygen in compound 'P' is \_\_\_\_\_ %
34. The total number of atoms in one molecule of compound 'T' is \_\_\_\_\_



## Mathematics

## PART – III

### SECTION – A

(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

35. Suppose function  $f(x)$  is continuous and differentiable at  $x = 0$ , and function  $g(x)$  is continuous but non-differentiable at  $x = 0$ . If the function  $h(x) = e^{g(x)} \cdot f^2(x) \cdot g(x)$  is differentiable at  $x = 0$ , then which of the following is CORRECT?  
 (A)  $f(0)$  must be equal to 0  
 (B)  $g(0)$  must be a particular negative value  
 (C)  $f(0) = 0$  or  $g(0) = -1$   
 (D)  $f(0) = g(0)$
36. Let  $d(P, A)$  denotes the distance between the point  $P$  and point  $A$ . If a variable point  $P$  moves such that  $d(P, (-4, 0)) + d(P, (0, 0)) \leq 6$  and  $d(P, (4, 0)) + d(P, (0, 0)) \geq 6$ , then the area of the region in which point  $P$  moves is equal to  
 (A)  $\frac{20}{3} + \sin^{-1} \frac{2}{3}$   
 (B)  $\frac{10}{3} + 6\sqrt{5} \sin^{-1} \left(\frac{2}{3}\right)$   
 (C)  $\frac{20}{3} + 6\sqrt{5} \sin^{-1} \frac{2}{3}$   
 (D)  $\frac{10}{3} + \sin^{-1} \left(\frac{2}{3}\right)$
37. If  $f_n(x) = \tan^{-1} \left\{ \left( \sum_{r=1}^n \operatorname{cosec} \frac{x}{2^{r-1}} \right) - \cot \frac{x}{2^n} \right\}$ , then which of the following is CORRECT?  
 (A)  $f_5 \left( \frac{\pi}{4} \right) = \frac{3\pi}{4}$   
 (B)  $f_5 \left( \frac{3\pi}{4} \right) = \frac{3\pi}{4}$   
 (C)  $f_5 \left( \frac{3\pi}{4} \right) = \frac{\pi}{4}$   
 (D)  $f_5 \left( \frac{\pi}{4} \right) = \frac{\pi}{4}$
38. Consider the function  $f(x) = \frac{\log_e(1+|x|)}{|x|}$ , then which of the following statement about  $f(x)$  is FALSE? (Where  $[.]$  represents greatest integer function)  
 (A)  $\lim_{x \rightarrow 0} [f(x)] = 0$   
 (B)  $\left[ \lim_{x \rightarrow 0} f(x) \right] = 1$   
 (C)  $f(x)$  is a bounded function  
 (D)  $f(x)$  has a point of minima

### SECTION – A

(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39.  $f(x) = \frac{x^2 - 3x + 2}{|x^2 - 3x + 2|}$  and  $g(x) = \operatorname{sgn}(x - 1)(x - 2)$ , then which of the following is/are TRUE?  
 (A)  $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} g(x)$   
 (B)  $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} g(x)$   
 (C)  $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} g(x)$   
 (D)  $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} g(x)$

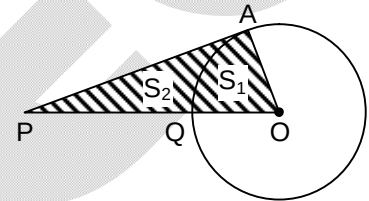
40. If a differentiable function  $f(x)$  satisfy the following relation  $f(x) + f(x^2) + f(x^3) = f(x + x^2 + x^3) \forall x \in \mathbb{R}$ , then which of the following is/are TRUE?  
 (A)  $f'(-1) = f'(3)$  (B)  $f'(-1) = f'(1)$   
 (C)  $f'(1) = f'(3)$  (D)  $3f(-1) = f(3)$
41. Which of the following(s) is/are CORRECT?  
 (A) If  $f(x)$  is periodic and  $g(x)$  is non-linear and non-periodic, then  $f(g(x))$  is non-periodic  
 (B) If  $f(x)$  is periodic and  $g(x)$  is non-linear and non-periodic, then  $f(g(x))$  can be periodic with a defined period  
 (C) If  $f(x)$  and  $g(x)$  both are non-periodic, then  $f(g(x))$  can be periodic  
 (D) If  $f(x)$  and  $g(x)$  both the non-periodic, then  $f(g(x))$  cannot be periodic with a defined period

### SECTION – B

#### (Numerical Answer Type)

This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

42. In the figure, OQ is fixed diameter of the circle, and AP is tangent to the circle which cut the line OQ at P. Let  $S_1$  is area of sector AOQ, and area of curvilinear triangle APQ is  $S_2$ . Then  $\lim_{A \rightarrow Q} \frac{S_2}{S_1}$  is equal to



43. A circle has center  $(0, 0)$  and variable radius  $r$ . If tangents are drawn from the point  $(-3, \sqrt{7})$  to the circle, then the value of  $r$  for which area of the triangle formed by the tangents and chord of contact is maximum, is equal to
44. The maximum value of the function  $f(x) = \frac{4x^2 - 4x + 1}{8x^4 - 16x^3 + 12x^2 - 4x + 1}$  is equal to
45. Consider the function  $f(x) = x\sqrt{1+(x+1)}\sqrt{1+(x+2)}\sqrt{1+(x+3)}\sqrt{1+\dots}$ , then  $\lim_{n \rightarrow \infty} \sum_{x=1}^n \frac{1}{f(x)}$  is  $\frac{p}{q}$  where  $p$  and  $q$  are coprime, then  $p + q$  is equal to
46. If  $\int \frac{1 + \frac{1}{2x^2} + \frac{1}{4x^4} + \dots}{x + \frac{1}{3x} + \frac{1}{5x^3} + \dots} dx = f(x) + C$  where  $C$  is integration constant. Then  $\lim_{x \rightarrow \infty} f(x)$  is equal to
47. Consider the differential equation  $\frac{dy}{dx} = \frac{1 - \ln x}{x^2 + \ln^2 x}$ . Given that  $y(1) = \frac{\pi}{2}$ , then  $\left[ y\left(\frac{1}{e}\right) \right]$  is equal to (Where  $[.]$  represents greatest integer function)

**SECTION – C**  
**(Numerical Answer Type)**

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

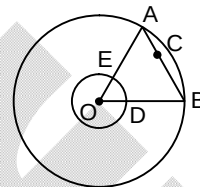
**Paragraph for Question Nos. 48 and 49**

There are two concentric circles (centred at point O) of radii 2 and 6 respectively, as shown in the figure. B is a fixed point while point A varies on the outer circle.

C is a point on AB such that  $AC : BC = 1 : 2$

E and D lies on inner circle and on OA and OB

Let S is the area of  $\triangle ECD$ . Then answer the following question:



48. Maximum possible area of  $\triangle ECD$  i.e.  $S_{\max}$  will be equal to
49. For  $S_{\max}$ , the common area between  $\triangle ECD$  and inner circle is equal to

**Paragraph for Question Nos. 50 and 51**

Consider the following functions  $f(x) = \sin x$  and  $g(x) = x - \sin x$  and answer the following questions:

50. If fundamental periods of  $f(g(x))$  and  $g(f(x))$  are  $T_1$  and  $T_2$ , then  $T_1 + T_2$  will be equal to

51.  $\int_0^{2\pi} (f(g(x)) + g(f(x))) dx$  is equal to